

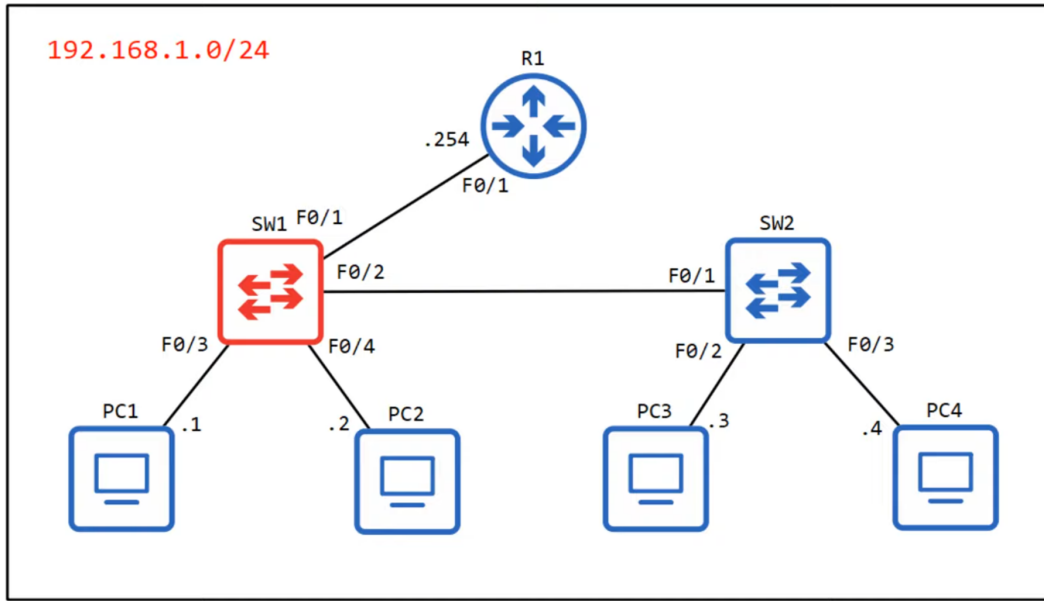
Day 9 - Switch Interfaces

Saturday, August 22, 2026

2:50 PM

9. SWITCH INTERFACES

Network Topology



CISCO CLI for SWITCHES

show ip interface brief

```
SW1>en
SW1#sh ip int br
Interface                IP-Address    OK? Method Status      Protocol
Vlan 1                   unassigned    YES unset  up          up
FastEthernet0/1          unassigned    YES unset  up          up
FastEthernet0/2          unassigned    YES unset  up          up
FastEthernet0/3          unassigned    YES unset  up          up
FastEthernet0/4          unassigned    YES unset  up          up
FastEthernet0/5          unassigned    YES unset  down        down
FastEthernet0/6          unassigned    YES unset  down        down
FastEthernet0/7          unassigned    YES unset  down        down
FastEthernet0/8          unassigned    YES unset  down        down
FastEthernet0/9          unassigned    YES unset  down        down
FastEthernet0/10         unassigned    YES unset  down        down
FastEthernet0/11         unassigned    YES unset  down        down
FastEthernet0/12         unassigned    YES unset  down        down
```

// enter Privileged EXEC mode

SW1>enable

// Show all interfaces of Switch 1.

SW# show ip interface brief

This will show the interfaces currently on Switch 1. It has the same information structure as Cisco Routers.

Notice the Status (Layer 2) and Protocol (Layer 1) columns are showing "up/up".

Unlike ROUTERS, SWITCHES do no DEFAULT to 'administrative down/down'(shutdown).

Unconnected devices will show as "down" and "down" (not connected to another device)



show interfaces status

```
SW1#show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	1	a-full	a-100	10/100BaseTX
Fa0/2		connected	trunk	a-full	a-100	10/100BaseTX
Fa0/3		connected	1	a-full	a-100	10/100BaseTX
Fa0/4		connected	1	a-full	a-100	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX

// Show the status of all interfaces on SW1

SW1#show interfaces status

This will list:

- Ports
- Name (which is description)
- Status (connection status)
- Vlan (can be used to divide up LANs) - Vlan 1 is the default.
- Duplex (can the connection send/receive at same time?) - Auto is default
- Speed (speed in bps) - Auto is default
- Type (what medium is being used, speed of interface)

Config commands

speed

duplex

description



Configuring interface speed and duplex

```
SW1#conf t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
SW1(config)#int f0/1
```

```
SW1(config-if)#speed ?
```

```
10 Force 10 Mbps operation
```

```
100 Force 100 Mbps operation
```

```
auto Enable AUTO speed configuration
```

```
SW1(config-if)#speed 100
```

```

SW1(config-if)#speed 100
SW1(config-if)#duplex ?
auto          Enable AUTO duplex configuration
full          Force full duplex operation
half          Force half-duplex operation
SW1(config-if)#duplex full
SW1(config-if)#description ## to R1 ##

```

Configuring interface speed and duplex

```

SW1#sh int status
Port    Name          Status      Vlan    Duplex  Speed  Type
Fa0/1    ## to R1 ##    connected   1        full    100    10/100BaseTX
Fa0/2    ## to R1 ##    connected   trunk    a-full  a-100  10/100BaseTX
Fa0/3    ## to R1 ##    connected   1        a-full  a-100  10/100BaseTX
Fa0/4    ## to R1 ##    connected   1        a-full  a-100  10/100BaseTX
Fa0/5    ## to R1 ##    notconnect  1        auto    auto   10/100BaseTX
Fa0/6    ## to R1 ##    notconnect  1        auto    auto   10/100BaseTX
Fa0/7    ## to R1 ##    notconnect  1        auto    auto   10/100BaseTX
Fa0/8    ## to R1 ##    notconnect  1        auto    auto   10/100BaseTX
Fa0/9    ## to R1 ##    notconnect  1        auto    auto   10/100BaseTX
Fa0/10   ## to R1 ##    notconnect  1        auto    auto   10/100BaseTX
Fa0/11   ## to R1 ##    notconnect  1        auto    auto   10/100BaseTX
Fa0/12   ## to R1 ##    notconnect  1        auto    auto   10/100BaseTX

```

INTERFACE RANGE

Unused Interfaces can pose a security risk so it's a good idea to deactivate them.

However, if you have 28+ interfaces not in use, do you have to do them one at a time?

Answer: No! There is a command to apply configurations to a range of interfaces.

Inside Global Config Mode (config t):

```

interface range
SW1(config)#interface range f0/5 - 12
SW1(config-if-range)#description ## not in use ##
SW1(config-if-range)#shutdown
00:42:36: %LINK-5-CHANGED: Interface FastEthernet0/5, changed state to administratively down
00:42:36: %LINK-5-CHANGED: Interface FastEthernet0/6, changed state to administratively down
00:42:36: %LINK-5-CHANGED: Interface FastEthernet0/7, changed state to administratively down
00:42:36: %LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down
00:42:36: %LINK-5-CHANGED: Interface FastEthernet0/9, changed state to administratively down
00:42:36: %LINK-5-CHANGED: Interface FastEthernet0/10, changed state to administratively down
00:42:36: %LINK-5-CHANGED: Interface FastEthernet0/11, changed state to administratively down
00:42:36: %LINK-5-CHANGED: Interface FastEthernet0/12, changed state to administratively down
SW1(config-if-range)#

```

SW1(config)#interface range f0/5 - 12 // Choose all interfaces from 0/5 to 0/12

SW1(config-if-range)#description ## not in use ##

SW1(config-if-range)#shutdown

<< this will list all the interfaces being set to administratively down >>

Confirm with 'show interface status' in Privileged EXEC mode or if in CONFIG mode, use 'do show interface status'

Configuring switch interfaces

```

SW1(config-if-range)#do sh int status
Port    Name          Status      Vlan    Duplex  Speed  Type
Fa0/1    ## to R1 ##    connected   1        full    100    10/100BaseTX
Fa0/2    ## to R1 ##    connected   trunk    a-full  a-100  10/100BaseTX
Fa0/3    ## to R1 ##    connected   1        a-full  a-100  10/100BaseTX

```

```

Fa0/3  ## to end hosts ## connected 1 a-full a-100 10/100BaseTX
Fa0/4  ## to end hosts ## connected 1 a-full a-100 10/100BaseTX
Fa0/5  ## not in use ## disabled 1 auto auto 10/100BaseTX
Fa0/6  ## not in use ## disabled 1 auto auto 10/100BaseTX
Fa0/7  ## not in use ## disabled 1 auto auto 10/100BaseTX
Fa0/8  ## not in use ## disabled 1 auto auto 10/100BaseTX
Fa0/9  ## not in use ## disabled 1 auto auto 10/100BaseTX
Fa0/10 ## not in use ## disabled 1 auto auto 10/100BaseTX
Fa0/11 ## not in use ## disabled 1 auto auto 10/100BaseTX
Fa0/12 ## not in use ## disabled 1 auto auto 10/100BaseTX

```

FULL / HALF DUPLEX

HALF DUPLEX:

- Device cannot send / receive data at the same time. If it is receiving a frame, it must wait before sending a frame.

FULL DUPLEX:

- Device CAN send / receive data at the same time. It does NOT have to wait.

MOST modern SWITCHES support FULL DUPLEX.

WHERE is HALF DUPLEX used? Almost nowhere.

In the past, LAN HUBS used HALF DUPLEX.

When multiple packets were received by the HUB, the HUB would simply FLOOD the connections with frame data, causing a COLLISION (on the interface), and hosts would not receive the frame intact.

All devices connected to a HUB are called a COLLISION DOMAIN.

To DEAL with COLLISIONS, Ethernet devices use a mechanism called CSMA/CD.

CSMA/CD = CARRIER SENSE MULTIPLE ACCESS with COLLISION DETECTION.

- Before sending frames, devices 'listen' to the collision domain until they detect that other devices are not sending.
- If a collision occurs, the device sends a jamming signal to inform the other devices that a collision happened.
- Each device will wait a random period of time before sending frames again.
- The process repeats.

SWITCHES are more sophisticated than HUBS.

HUBS are Layer 1 Devices - Collisions are common and use CSMA/CD. SWITCHES are Layer 2 Devices - Collisions RARELY occur.



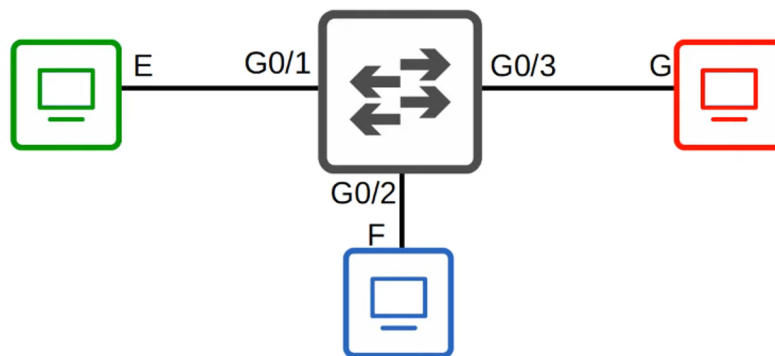
Speed/Duplex Autonegotiation

- Interfaces that can run at different speeds (10/100 or 10/100/1000) have default settings of speed auto and duplex auto.
- Interfaces 'advertise' their capabilities to the neighboring device, and they negotiate the best speed and duplex settings they are both capable of.

E = 10

F = 10/100

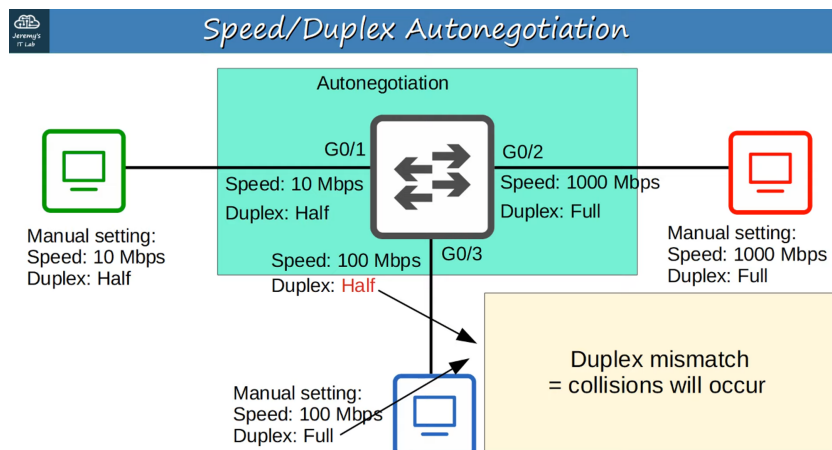
G = 10/100/1000



SPEED / DUPLEX AUTONEGOTIATION

- Interfaces that can run at different speeds (10/100 or 10/100/1000) have a default setting of SPEED AUTO and DUPLEX AUTO.
- Interfaces 'advertise' their capabilities to the neighbouring device, and they negotiate the best SPEED and DUPLEX settings they are both capable of.

WHAT if AUTONEGOTIATION is DISABLED on the device connected to the SWITCH ?



- **SPEED:** The SWITCH will try to send at the speed that the other device is operating at. If it fails to send the speed, it will use the slowest supported speed (ie: 10 Mbps on a 10/100/1000 interface).
- **DUPLEX:** If the speed is 10 or 100 Mbps the SWITCH will use HALF DUPLEX. If the speed is 1000 Mbps or great, it will use FULL DUPLEX.

INTERFACE COUNTERS AND ERRORS

Show using the:

// Privileged EXEC mode

SW1#show interfaces

Error stats will be at the bottom.

Interface Errors

```
SW1#show interfaces f0/2
FastEthernet0/2 is up, line protocol is up
Hardware is Fast Ethernet, address is 000C.3168.8461 (bia 000C.3168.8461)
Description: ## to SW2 ##
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
```

```
reliability 255/255, txload 1/255, rxload 1/255
Auto-duplex, Auto-speed
Encapsulation ARPA, loopback not set
ARP type: ARPA, ARP Timeout 04:00:00
Last input 02:29:44, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queuing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  269 packets input, 71059 bytes, 0 no buffer
    Received 6 broadcasts, 0 runts, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
  7290 packets output, 429075 bytes, 0 underruns
  0 output errors, 3 interface resets
  0 output buffer failures, 0 output buffers swapped out
```

Packets Received / Total bytes received.

Runts: Frames that are smaller than the minimum frame size (64 bytes)

Giants: Frames that are larger than the maximum frame size (1518 bytes)

CRC: Frames that failed the CRC check (in the Ethernet FCS trailer)

Frame: Frames that have an incorrect format (due to an error)

Input errors: Total of various counters, such as the above four

Output errors: Frames the SWITCH tried to send, but failed due to an error